

PROJECT PROPOSAL

Auction Platform

An eBay-style online auction and marketplace system

Web platform • Admin back-office • 4-month delivery

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Version	1.0
Project fee	NPR 13,00,000 excluding VAT
Duration	4 months (16 weeks)

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1. Abstract

This proposal describes the design and development of a web-based **online auction and marketplace platform** that follows the proven workflows of eBay: sellers list items for auction, at a fixed price or with Best Offer; buyers search, bid, buy and pay; the platform protects both sides through escrow, feedback, returns and dispute resolution.

The platform is built as a **modular monolith** with three runtime processes (API, real-time gateway and background worker), backed by PostgreSQL, Redis, OpenSearch and object storage. Money is handled through a **double-entry ledger** with escrow, and auctions use **proxy bidding** with server-side locking so that concurrent bids are always processed correctly.

The scope covers the web storefront and the admin back-office. Native mobile apps are not included. The work will be delivered in **four months** for a fixed fee of **NPR 13,00,000 excluding VAT**. Servers, storage and payment accounts are provided by the client.

At a glance

Delivery: 4 months, 4 milestones | Fee: NPR 13,00,000 excluding VAT | Platform: web only

Client provides: servers, storage, payment gateway accounts, SMS and email services

2. Introduction

2.1 Background

Online auctions combine discovery, competitive pricing and trust between strangers. A working auction marketplace needs much more than a listing page: a bidding engine that stays correct under simultaneous bids, a payment flow that holds funds safely, and back-office tools to resolve problems. The client has asked for a system that offers the same functionality as eBay. Because eBay is the product of thousands of engineers, this proposal defines a focused, deliverable release in four months and lists the remaining eBay features as an optional Phase 2.

2.2 Objectives

- Let any registered user sell and buy through auction, fixed price and Best Offer listings.
- Provide a fair, real-time bidding experience with automatic proxy bidding.
- Protect buyers and sellers with escrow, returns, disputes and reputation scores.
- Give administrators full control of users, catalog, money and moderation.
- Build on a clean architecture that can grow (new features, higher traffic) without a rewrite.

2.3 Scope of this release

All features marked **Included** in section 4, delivered as a responsive website (desktop, tablet and phone browsers) plus the admin panel. Integration with up to two payment gateways, one SMS provider, one email provider, and manual courier tracking entry.

2.4 Out of scope (optional Phase 2)

Features marked **Phase 2** in section 4, native iOS and Android apps, machine-learning ranking and fraud models, image search, multi-region hosting, and any work not described in this document. Phase 2 items can be quoted separately.

2.5 Assumptions

- The client nominates one decision-maker who reviews work and approves milestones within 5 working days.
- Server, storage and payment accounts are ready on the dates in section 17.
- The platform operates in Nepal in NPR; the interface is in English (Nepali translation is Phase 2).
- Business rules (fee rates, policies, categories) are supplied by the client during month 1.

3. User roles

Role	What they can do
Guest	Browse categories, search, view listings and seller profiles; must register to bid or buy
Buyer	Bid, buy, make offers, pay, track orders, message sellers, return items, open cases, leave feedback, use watchlist and saved searches
Seller	Everything a buyer can do, plus complete KYC, create and manage listings, ship orders, handle offers, returns and cases, view dashboard and payouts
Support agent	Handle tickets, moderate listings and messages, work the dispute queue
Finance officer	Review fees, approve payouts and refunds, view ledger and reconciliation reports
Administrator	Full control: users, roles, catalog, settings, content and audit log

4. Features

Every feature of the platform is listed below by module. The **Release** column shows whether the feature is delivered within the four-month, fixed-fee scope (**Included**) or is available as an optional later phase (**Phase 2**).

4.1 User and account management

Feature	Details	Release
Registration and login	Email or phone with password; OTP verification; Google sign-in	Included
Two-factor authentication	OTP or authenticator app; mandatory for staff and sellers	Included
Password recovery and sessions	Reset by email or OTP; list and revoke active devices	Included
Roles and account types	Buyer and seller on one account; individual or business account	Included
Profile and address book	Personal and business profile, multiple addresses, defaults	Included
Seller KYC verification	ID and address proof upload, admin review, approve or reject with reason	Included
Feedback score and seller levels	Positive/neutral/negative counts; new, standard, above standard, top rated	Included
Restriction, suspension and appeals	Admin actions with reason, duration and appeal workflow	Included

4.2 Catalog and categories

Feature	Details	Release
Category tree	Unlimited levels, slug, icon, banner, ordering, activation	Included

Feature	Details	Release
Item specifics builder	Per-category attributes: text, dropdown, multi-select, number, date, yes/no	Included
Conditions and brands	New, used, refurbished, for parts; verified and restricted brands	Included
Category rules	Allowed formats, approval requirement, minimum photos, commission rate	Included
Restricted and prohibited items	Rule text and document requirements per category	Included

4.3 Listings

Feature	Details	Release
Create and edit listing	Title, category, condition, item specifics, photos, rich description	Included
Listing formats	Auction, Buy It Now (fixed price) and Best Offer	Included
Variations and quantity	Size/colour combinations with own price, stock and photo	Included
Photos	Multiple images, drag-to-order, automatic resize, moderation status	Included
Drafts, scheduling, relist, sell similar	Save draft, start later, auto-relist, copy an existing listing	Included
Bulk upload	CSV import with error report	Included
Listing fees	Insertion and final value fees per category and seller level	Included
Moderation and reporting	Approval queue, report listing, remove or suspend	Included
Listing templates and upgrades	Reusable templates; bold, highlight, featured placements	Phase 2
Video on listings	Short video upload and playback	Phase 2

4.4 Auction and bidding engine

Feature	Details	Release
Proxy (automatic) bidding	System bids up to the buyer's maximum in fixed increments	Included
Starting price, reserve, increments	Configurable increment table by price range	Included
Duration and soft close	Auction length; automatic extension if a bid arrives in the last minutes	Included
Live updates	Real-time price, bid count and countdown over WebSocket	Included
Bid rules	Bid retraction limits, blocked bidders, minimum feedback and payment requirements	Included
Winner and payment deadline	Winner determination, order creation, payment deadline, unpaid strikes	Included
Second-chance offer	Offer to next highest bidder if the winner does not pay	Included
Alerts	Outbid, ending soon, won, lost, reserve met	Included

4.5 Best Offer

Feature	Details	Release
Offers and counter-offers	Buyer offer, seller accept, decline or counter, limited rounds	Included
Auto rules	Auto-accept and auto-decline thresholds, offer expiry	Included

4.6 Search and discovery

Feature	Details	Release
Full-text search	Typo tolerance, synonyms, autocomplete suggestions	Included
Filters and sorting	Price, condition, format, location, free shipping, seller rating, item specifics; best match, price, ending soonest, newest	Included
Best Match ranking	Relevance combined with seller quality, sales velocity and freshness	Included
Watchlist, saved searches, recently viewed	Alerts on price drop and before auction end	Included
Personalised recommendations and trending	Similar items, customers also viewed, trending	Phase 2

4.7 Cart and checkout

Feature	Details	Release
Cart and checkout	Multi-seller cart, address selection, shipping choice, order summary	Included
Stock reservation	Items held for 15 minutes during checkout	Included
Coupons and discounts	Percentage, fixed or free-shipping coupons with limits	Included
VAT and shipping calculation	Tax rules by category; shipping by weight and zone	Included
Guest checkout	Purchase without registering	Phase 2

4.8 Payments, escrow and payouts

Feature	Details	Release
Payment gateway integration	Up to two client-provided gateways; hosted payment page; verified webhooks	Included
Escrow hold	Buyer funds held until delivery is confirmed or the protection window ends	Included
Double-entry ledger	Every money movement recorded as balanced, insert-only entries	Included
Fee engine	Commission and final value fee by category and seller level	Included
Seller wallet and payouts	Balance, payout account, payout request, admin approval, payout history	Included
Refunds	Full and partial refunds linked to returns and disputes	Included
Invoices and statements	VAT invoice per order; seller earnings statement	Included
Reconciliation	Gateway report matching and daily ledger integrity check	Included
Multi-currency	Additional currencies with exchange rates	Phase 2

4.9 Orders and shipping

Feature	Details	Release
Order lifecycle	Pending payment, paid, processing, shipped, delivered, completed, cancelled	Included
Shipping policies	Handling time, services, free-shipping threshold, excluded regions	Included
Tracking	Seller enters carrier and tracking number; delivery confirmation by buyer	Included

Feature	Details	Release
Cancellations and auto-complete	Cancellation requests; automatic completion after protection window	Included
Courier API and label printing	Automatic rates, labels and tracking updates from courier APIs	Phase 2
Cash on delivery and international shipping	COD collection tracking; customs details	Phase 2

4.10 Returns and disputes

Feature	Details	Release
Return requests	Reason, photos, requested resolution, seller approval, return tracking	Included
Cases	Item not received and item not as described, with response deadlines	Included
Evidence and decision	File uploads, message thread, admin decision, refund or side with seller	Included
Escalation and appeals	Automatic escalation on seller silence; one appeal round	Included
Buyer protection window	Configurable number of days after delivery	Included

4.11 Feedback and reputation

Feature	Details	Release
Feedback and ratings	Positive, neutral or negative with comment; seller reply	Included
Detailed seller ratings	Item as described, communication, shipping speed, shipping cost	Included
Seller performance metrics	Defect rate, late shipment rate, case rate, monthly level evaluation	Included

4.12 Messaging and notifications

Feature	Details	Release
Buyer–seller messaging	Conversation per listing or order; read status; attachments	Included
Message filters	Keyword-based spam and contact-sharing filter	Included
Email, SMS and in-app notifications	Template-based, per-event preferences, quiet hours	Included
Browser push notifications	Web push for bids and messages	Phase 2

4.13 Seller tools

Feature	Details	Release
Seller dashboard	Sales, orders, traffic, payouts, performance level	Included
Inventory and order tools	Bulk edit price and quantity, low-stock alert, order filters	Included
Reports	Sales, fees, payouts, returns; export to CSV	Included
Vacation mode and saved replies	Pause listings; reusable message templates	Included
Branded stores and subscriptions	Seller storefront, store plans and fee discounts	Phase 2

Feature	Details	Release
Seller promotions and promoted listings	Markdown sales, coupons, paid ad campaigns	Phase 2
Sub-accounts	Staff logins for sellers with limited permissions	Phase 2

4.14 Admin back-office

Feature	Details	Release
Role-based access	Roles and module-level permissions for staff	Included
Management consoles	Users, KYC, listings, categories and attributes, orders, disputes	Included
Finance console	Fees, payouts, refunds, ledger, tax settings, revenue reports	Included
Moderation queue	Flagged listings, images, messages; strikes and restrictions	Included
Content management	Banners, static pages, FAQs, help articles	Included
Settings and audit log	Site settings, auction rules, fee rules; full audit trail of admin actions	Included
Reports and analytics	GMV, users, listings, conversion, revenue, seller and category performance	Included
Support tickets	Basic ticket tracking with assignment and status	Included
Marketing campaigns	Email and push broadcasts, referral programme	Phase 2

4.15 Trust, safety and platform

Feature	Details	Release
Rule-based fraud checks	Velocity limits, blacklists, device fingerprint, risk score with review queue	Included
Strikes and reporting	Unpaid-item strikes; report user or listing	Included
Responsive web and SEO	Mobile-friendly web, clean URLs, sitemap, structured data	Included
Security and audit	Encryption, rate limiting, audit logs, automated backups, monitoring	Included
Advanced fraud (ML) and shill-bid detection	Model-based scoring; linked-account bidding patterns	Phase 2
Brand-owner claims, multi-language	Rights-owner takedown workflow; Nepali interface translation	Phase 2

5. Data models and fields

The tables below list every data model (database table) and its fields. Money fields end in **_minor** and hold whole numbers in the smallest currency unit (paisa). Fields marked (JSONB) hold structured data such as item specifics. Every table also carries **created_at** and **updated_at**; user-facing tables additionally carry **deleted_at** for soft deletion.

5.1 Identity

Table	Fields
users	id, public_id, username, email, phone, password_hash, account_type, status, seller_level, email_verified_at, phone_verified_at, created_at, updated_at, deleted_at
user_profiles	user_id, first_name, last_name, display_name, avatar_url, date_of_birth, language, currency, timezone, bio

Table	Fields
business_profiles	user_id, business_name, registration_no, pan_vat_no, business_type, address, website, contact_person
addresses	id, user_id, label, recipient_name, phone, line1, line2, city, province, postal_code, country, is_default_shipping, is_default_billing
kyc_documents	id, user_id, doc_type, doc_number (encrypted), front_url, back_url, selfie_url, status, rejection_reason, reviewed_by, reviewed_at
verification_codes	id, user_id, channel, code_hash, expires_at, attempts, verified_at
auth_sessions	id, user_id, refresh_token_hash, device, ip, user_agent, last_active_at, revoked_at
two_factor	user_id, method, secret (encrypted), backup_codes_hash, enabled_at
social_accounts	id, user_id, provider, provider_user_id, linked_at
account_restrictions	id, user_id, type, reason, starts_at, ends_at, appeal_status
user_blocks	blocker_id, blocked_id, reason, created_at

5.2 Catalog

Table	Fields
categories	id, parent_id, name, slug, level, path, icon_url, sort_order, commission_bp, allow_auction, allow_fixed, allow_offer, requires_approval, is_active, seo_title, seo_description
category_attributes	id, category_id, name, input_type, is_required, is_filterable, is_searchable, unit, sort_order
attribute_values	id, attribute_id, value, sort_order
conditions	id, name, description
brands	id, name, logo_url, is_verified, is_restricted
restricted_rules	id, category_id, rule_type, rule_text, requires_documents

5.3 Listings

Table	Fields
listings	id, public_id, seller_id, category_id, title, subtitle, description, condition_id, listing_type, status, currency, buy_now_price_minor, start_price_minor, reserve_price_minor, quantity, quantity_available, item_country, item_city, start_at, end_at, duration_days, relist_count, view_count, watch_count, attributes (JSONB), version, created_at, updated_at
listing_media	id, listing_id, url, thumb_url, sort_order, is_primary, width, height, file_size, moderation_status
listing_variations	id, listing_id, sku, attribute_combo, price_minor, quantity, media_id
listing_policies	listing_id, handling_days, free_shipping, shipping_cost_minor, ship_to_countries, returns_accepted, return_window_days, return_shipping_paid_by
listing_drafts	id, seller_id, data (JSONB), saved_at
listing_revisions	id, listing_id, changed_fields (JSONB), changed_by, changed_at
listing_fees	id, listing_id, fee_type, amount_minor, charged_at
listing_reports	id, listing_id, reporter_id, reason, details, status, resolved_by
bulk_uploads	id, seller_id, file_url, total_rows, success_count, error_count, error_report_url, status

5.4 Auctions and offers

Table	Fields
auctions	listing_id, start_price_minor, current_price_minor, reserve_price_minor, reserve_met, leader_id, leader_max_minor, bid_count, increment_rule_id, end_at, extension_count, status, winner_id, winning_bid_id
bids	id, listing_id, bidder_id, amount_minor, max_minor, bid_type, status, placed_at, ip, device_id
increment_rules	id, price_from_minor, price_to_minor, increment_minor
bid_retractions	id, bid_id, reason, requested_at, approved
blocked_bidders	seller_id, blocked_user_id, reason
second_chance_offers	id, listing_id, user_id, price_minor, expires_at, status
unpaid_strikes	id, user_id, order_id, created_at, expires_at
offers	id, listing_id, variation_id, buyer_id, seller_id, quantity, price_minor, message, status, expires_at, created_at
offer_counters	id, offer_id, from_role, price_minor, message, created_at
offer_rules	listing_id, auto_accept_minor, auto_decline_minor, max_counters

5.5 Search, cart and checkout

Table	Fields
search document (OpenSearch)	listing_id, title, description, category_path, attributes, condition, price, currency, listing_type, end_at, seller_id, seller_score, location, free_shipping, image_url, sold_count, popularity_score
saved_searches	id, user_id, query (JSONB), alert_frequency, channel, last_run_at
watchlist_items	id, user_id, listing_id, alert_on_price_drop, alert_before_end, added_at
recently_viewed	user_id, listing_id, viewed_at
search_synonyms	id, term, synonyms, language
carts	id, user_id, currency, created_at, updated_at, expires_at
cart_items	id, cart_id, listing_id, variation_id, seller_id, quantity, unit_price_minor, added_at
stock_reservations	id, listing_id, variation_id, user_id, quantity, expires_at
coupons	id, code, type, value, min_order_minor, max_discount_minor, usage_limit, per_user_limit, valid_from, valid_to, funded_by
coupon_usages	id, coupon_id, user_id, order_id, used_at
tax_rules	id, country, category_id, rate_bp, tax_type, is_inclusive, effective_from
shipping_rates	id, service_id, zone_from, zone_to, weight_from_g, weight_to_g, rate_minor

5.6 Payments, escrow and ledger

Table	Fields
payments	id, order_id, buyer_id, amount_minor, currency, method, gateway, gateway_txn_id, status, failure_reason, paid_at
payment_webhooks	id, gateway, event_id, payload, signature_valid, processed_at
ledger_accounts	id, owner_type, owner_id, currency, type
ledger_entries	id, txn_id, account_id, direction, amount_minor, ref_type, ref_id, created_at
escrow_holds	id, order_id, amount_minor, held_at, release_condition, release_at, released_at, status

Table	Fields
seller_wallets	id, seller_id, balance_minor, hold_minor, currency
payout_accounts	id, seller_id, type, bank_name, account_name, account_number (encrypted), branch, verified
payouts	id, seller_id, amount_minor, fee_minor, net_minor, status, initiated_at, completed_at, reference
fee_rules	id, category_id, seller_level, fee_type, percent_bp, fixed_minor, cap_minor, effective_from
order_fees	id, order_id, fee_type, amount_minor, rule_id
refunds	id, order_id, payment_id, amount_minor, reason, initiated_by, status, processed_at
invoices	id, order_id, invoice_no, issued_at, line_items (JSONB), tax_breakdown (JSONB), total_minor, pdf_url

5.7 Orders, shipping, returns and disputes

Table	Fields
orders	id, order_no, buyer_id, seller_id, source (auction / fixed / offer), status, payment_status, subtotal_minor, shipping_minor, tax_minor, discount_minor, total_minor, currency, placed_at, paid_at, shipped_at, delivered_at, completed_at, cancelled_at, cancel_reason
order_items	id, order_id, listing_id, variation_id, title_snapshot, sku, unit_price_minor, quantity, item_tax_minor, attributes_snapshot (JSONB)
order_status_history	id, order_id, from_status, to_status, changed_by, note, at
order_addresses	order_id, type, snapshot (JSONB)
cancellations	id, order_id, requested_by, reason, status, seller_response
shipping_policies	id, seller_id, name, handling_days, services (JSONB), free_shipping_threshold_minor, excluded_regions, local_pickup
shipping_services	id, name, delivery_min_days, delivery_max_days, cost_type, base_cost_minor
shipments	id, order_id, carrier, service_id, tracking_number, weight_g, cost_minor, status, shipped_at, delivered_at
tracking_events	id, shipment_id, status, location, description, event_time
returns	id, order_id, item_id, buyer_id, seller_id, reason, detail, requested_resolution, status, requested_at
return_shipments	id, return_id, carrier, tracking_number, shipped_at, received_at
cases	id, order_id, buyer_id, seller_id, case_type, status, amount_claimed_minor, opened_at, seller_response_due, escalated_at, closed_at
case_messages	id, case_id, sender_id, role, body, attachments, sent_at
case_evidence	id, case_id, uploaded_by, file_url, type
case_decisions	id, case_id, decided_by, outcome, refund_minor, seller_penalty, reason, decided_at

5.8 Feedback, messaging and seller tools

Table	Fields
feedback	id, order_id, from_user_id, to_user_id, role, rating, comment, seller_reply, created_at
seller_ratings	feedback_id, item_as_described, communication, shipping_speed, shipping_cost
seller_metrics	seller_id, period, defect_rate, late_shipment_rate, unresolved_cases_rate, tracking_upload_rate, total_sales

Table	Fields
conversations	id, buyer_id, seller_id, listing_id, order_id, last_message_at, status
messages	id, conversation_id, sender_id, body, attachments, is_read, read_at, flagged, sent_at
notifications	id, user_id, type, title, body, data (JSONB), channel, is_read, status, sent_at
notification_templates	id, event, channel, subject, body, language
notification_preferences	user_id, event_type, channels, quiet_hours
seller_dashboard_stats	seller_id, date, sales_minor, orders, views, conversion_rate
vacation_mode	seller_id, starts_at, ends_at, message, hide_listings
saved_replies	id, seller_id, title, body

5.9 Admin, trust and system tables

Table	Fields
admin_users	id, name, email, role_id, status, last_login_at
roles / permissions	roles(id, name); permissions(id, role_id, module, action)
audit_logs	id, actor_id, action, entity_type, entity_id, old_value, new_value, ip, at
cms_pages / banners	cms_pages(id, slug, title, body, status); banners(id, image_url, link, position, starts_at, ends_at)
site_settings	key, value, updated_by, updated_at
moderation_queue	id, item_type, item_id, reason, score, assigned_to, decision, decided_at
support_tickets	id, user_id, subject, category, priority, status, assigned_to, created_at
fraud_rules / risk_events	fraud_rules(id, name, condition, action, score); risk_events(id, user_id, event_type, score, details, at)
blacklists / device_fingerprints	blacklists(id, type, value, reason); device_fingerprints(id, user_id, device_hash, first_seen, last_seen)
strikes	id, user_id, type, reason, expires_at
outbox	id, aggregate, event_type, payload (JSONB), created_at, published_at
idempotency_keys	key, request_hash, response (JSONB), created_at

6. System architecture

6.1 Architecture style

The system is a **modular monolith**: one codebase organised into independent domain modules, deployed as three processes. This is faster to build and cheaper to run than microservices, and each module can be split into its own service later if traffic requires it.

6.2 Runtime components

Component	Responsibility	Technology
Storefront	Public pages, SEO, buyer and seller account screens	Next.js (server-side rendering)
Admin app	Staff back-office, restricted by IP and 2FA	React
API process	All REST endpoints and business rules	NestJS (TypeScript)
Realtime process	Live bid price, countdown, chat and notifications	WebSocket with Redis pub/sub

Component	Responsibility	Technology
Worker process	Queue consumers and scheduled jobs	NestJS worker
PostgreSQL	System of record for all business data	PostgreSQL 16
Redis	Sessions, cache, rate limits, locks, live auction cache	Redis 7
OpenSearch	Listing search, autocomplete, filters	OpenSearch 2.x
Message queue	Domain events and background jobs	RabbitMQ
Object storage and CDN	Images, KYC files, invoices, backups	S3-compatible storage

6.3 Module map and dependency rules

Domain modules: identity, catalog, listings, auctions, offers, search, cart, checkout, payments (ledger, escrow, payouts), orders, shipping, returns, disputes, feedback, messaging, notifications, seller, admin and fraud. Rules that keep the design clean:

1. A module owns its tables. No other module queries them directly.
2. Modules talk only through another module's public service interface or through events.
3. Allowed direction of calls: checkout to cart, listings, payments, orders; auctions to listings, payments, fraud; orders to payments, shipping, notifications; disputes to orders, payments, messaging.
4. Search, notifications, feedback and admin consume events; nothing depends on them.
5. No circular dependencies; enforced automatically in the build.
6. Every state change that others care about writes an event to the outbox table in the same database transaction.

6.4 Technology stack

Layer	Choice
Backend	Node.js with NestJS (TypeScript)
Web frontend	Next.js (React) for storefront; React admin panel
Database	PostgreSQL 16 with JSONB
Cache and locks	Redis
Search	OpenSearch
Queue	RabbitMQ
Real-time	WebSocket (Socket.IO)
Files	S3-compatible object storage behind a CDN
Containers and delivery	Docker, GitHub Actions, Terraform or shell scripts for provisioning
Monitoring	Prometheus and Grafana, Sentry for errors, central log store

7. Development architecture

7.1 Layers inside every module

Each module has the same four layers. Calls only go downward, which keeps business rules testable and the database hidden behind a single layer.

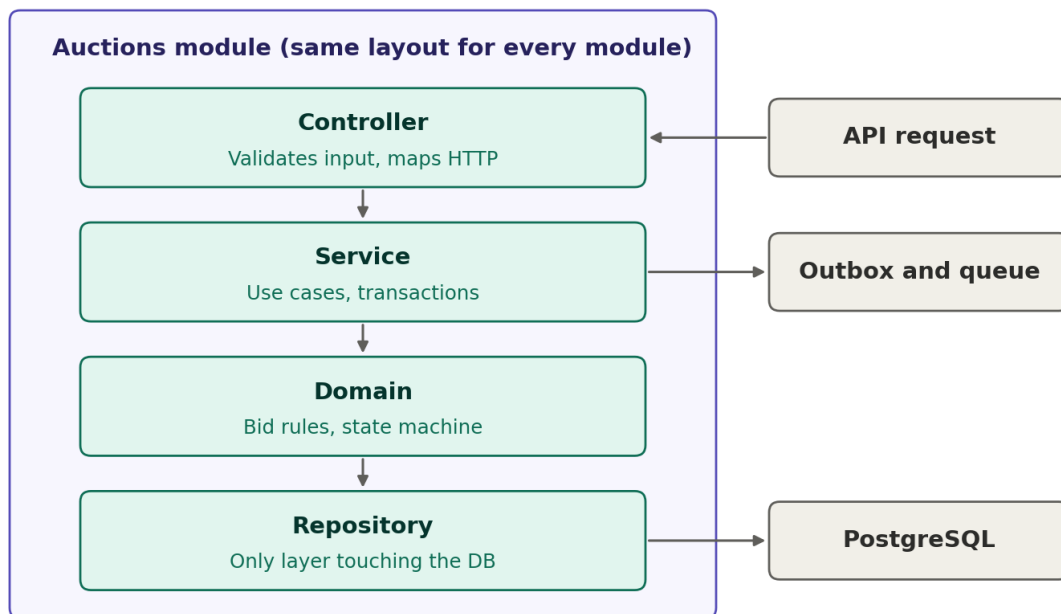


Figure 1: Layers inside one module

- **Controller**: validates input, maps HTTP to use cases. No business logic.
- **Service**: runs one use case in one database transaction and writes outbox events.
- **Domain**: pure business rules such as proxy-bid calculation and order state machine; no framework code.
- **Repository**: the only layer that talks to the database.

7.2 Repository structure

```
marketplace/
  apps/
    api/      modules: identity, catalog, listings, auctions, offers, search,
              cart, checkout, payments, orders, shipping, returns, disputes,
              feedback, messaging, notifications, seller, admin, fraud
    realtime/ WebSocket gateway for bids and chat
    worker/   queue consumers and scheduled jobs
    web/      Next.js storefront
    admin/    React admin panel
  packages/
    shared-types/ DTOs, enums, API client generated from OpenAPI
    ui/         shared design system components
  infra/      Docker, deployment scripts, environment templates
  docs/      OpenAPI, architecture decisions, runbooks, ER diagrams
  .github/    CI/CD workflows
```

7.3 Development practices

- **Contract first**: the OpenAPI specification is written before coding; frontend and backend share generated types.
- **Trunk-based development**: short-lived branches, pull-request review on every change, feature flags for unfinished work.
- **Database migrations**: versioned and backward compatible, so releases need no downtime.
- **Design system**: shared tokens (colour, type, spacing) and components used by the storefront and admin panel.
- **Environments**: local (Docker Compose), staging (production-like, masked data) and production.

7.4 Delivery pipeline and testing

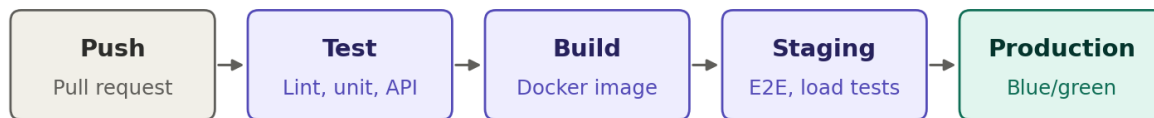


Figure 2: CI/CD pipeline with manual approval before production

Test type	Coverage
Unit tests	Domain rules: proxy bidding, fee calculation, order state machine, ledger balancing
Integration tests	API with a real database in containers; payment webhook handling, including duplicates
Concurrency tests	Many simultaneous bids on one auction; stock reservation under load
End-to-end tests	Register, list, bid, pay, ship, deliver, release escrow, feedback
Load tests	Search, bidding and checkout against the targets in section 10.3
Security tests	Automated dependency and secret scanning; OWASP Top 10 review

8. Database design

8.1 What is stored where

Data	Store
Users, listings, bids, offers, orders, payments, ledger, cases, feedback	PostgreSQL (source of truth)
Item specifics of each listing	PostgreSQL JSONB column, defined per category
Live auction price and leader (read cache)	Redis, backed by the PostgreSQL row
Search documents	OpenSearch, built from events and fully rebuildable
Sessions, rate-limit counters, locks	Redis
Images, video, KYC scans, invoices, dispute evidence	Object storage; metadata rows in PostgreSQL

8.2 Design rules

- Money is stored as integers in the smallest unit plus a currency code. Floating-point numbers are never used for money.
- Every table has a numeric primary key and a public UUID used in URLs.
- All timestamps are stored in UTC and shown in the user's time zone.
- Financial rows and bids are never deleted. User-facing content uses soft delete.
- Orders store snapshots of title, price and address, so later listing edits never change history.
- Frequently edited rows (listings, auctions, orders) carry a version number for optimistic locking.
- Status columns are constrained to allowed values; important state changes are recorded in history tables.
- Large append-only tables (bids, ledger entries, notifications, tracking events, audit logs) are partitioned by month.

8.3 Double-entry ledger

All money movement is recorded as balanced debit and credit entries grouped by a transaction ID. Entries are insert-only; a mistake is corrected with a reversing entry, never an edit. A nightly job checks that every transaction balances and that each account balance equals the sum of its entries.

Event	Debit	Credit
Buyer payment captured	gateway_clearing	escrow
Escrow released after delivery	escrow	seller_wallet (net), platform_revenue (fee), tax_payable
Refund to buyer	escrow	gateway_clearing
Payout to seller	seller_wallet	gateway_clearing

8.4 Indexing, scaling and backup

- Composite and partial indexes that match real queries, for example active listings ordered by end time.
- Connection pooling with PgBouncer; one primary and two read replicas for browsing and reports.
- Search and heavy filtering run on OpenSearch, never on the transactional database.
- Continuous log archiving with point-in-time recovery plus daily snapshots, stored in a separate location. Target recovery point 5 minutes, recovery time 1 hour.
- Financial and order records are retained according to Nepal's legal requirements; expired sessions and one-time codes are purged automatically.

9. ER diagrams

The entity relationship model is shown in six diagrams for readability. Lines show one-to-many relationships: a single bar marks the “one” side and a crow’s foot marks the “many” side; a small circle means the relation is optional. Grey headers mark tables that are shown in full in another diagram.

Diagram 1: Users and catalog

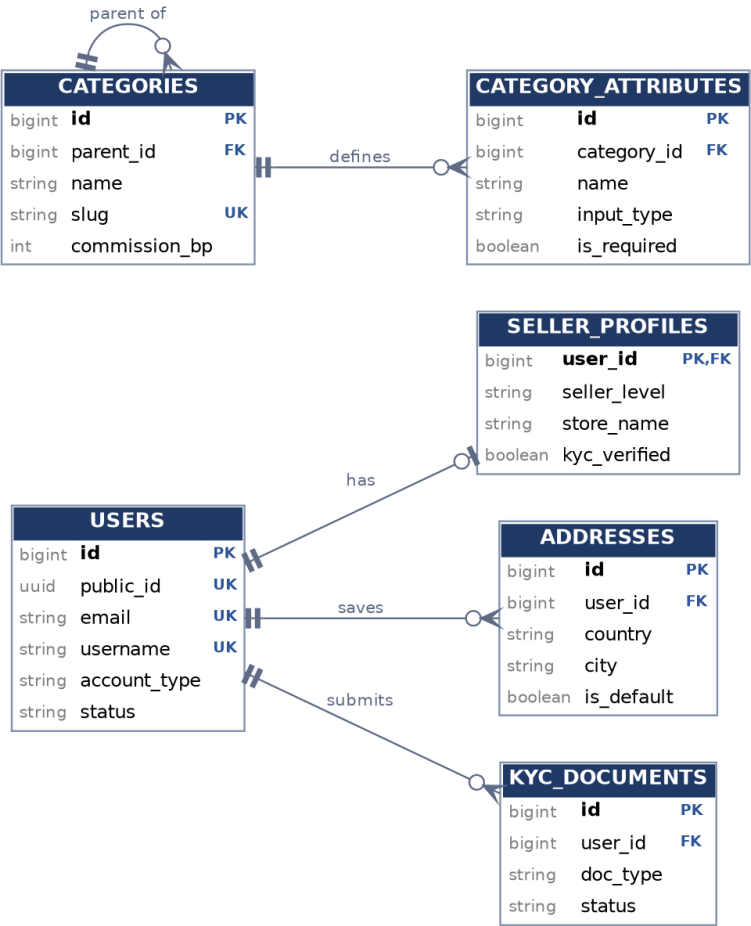


Figure 3: Users, seller profiles, addresses, KYC, categories

Diagram 2: Listings and auctions

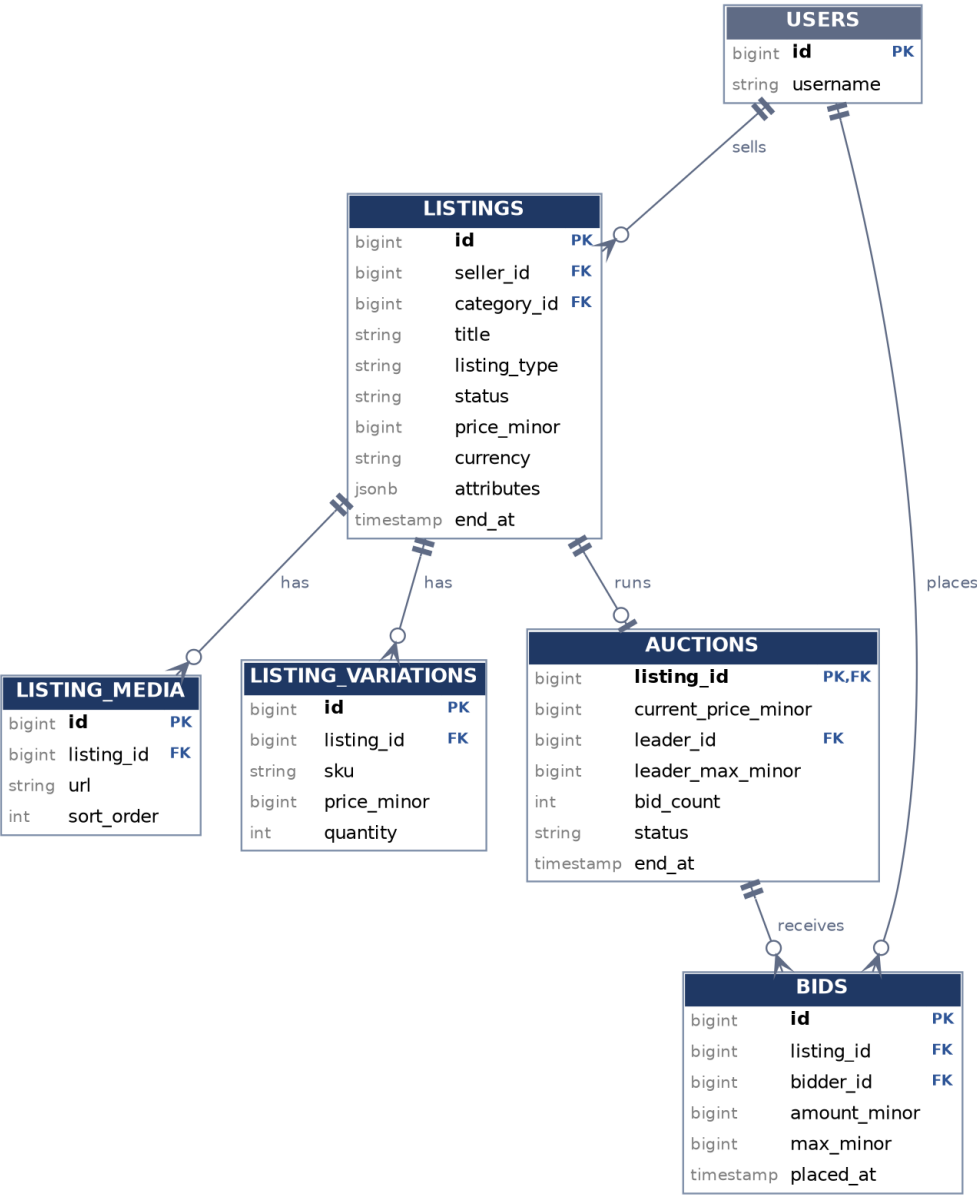


Figure 4: Listings, media, variations, auctions and bids

Diagram 3: Offers

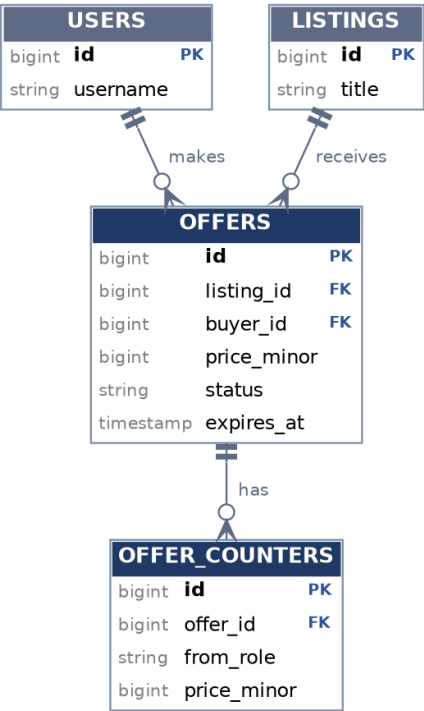


Figure 5: Best Offer and counter-offers

Diagram 4: Orders and fulfilment

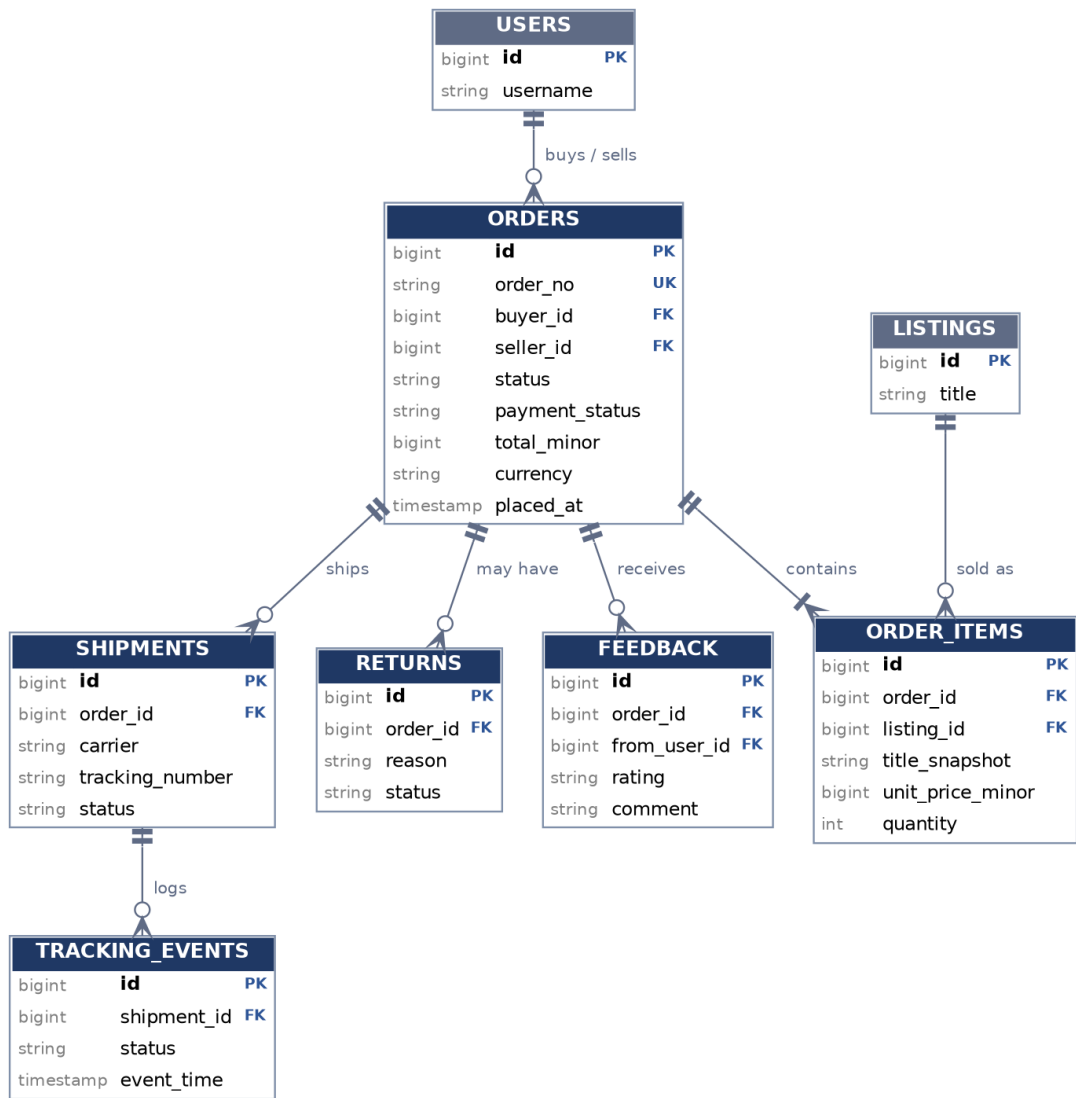


Figure 6: Orders, items, shipments, returns and feedback

Diagram 5: Payments and payouts

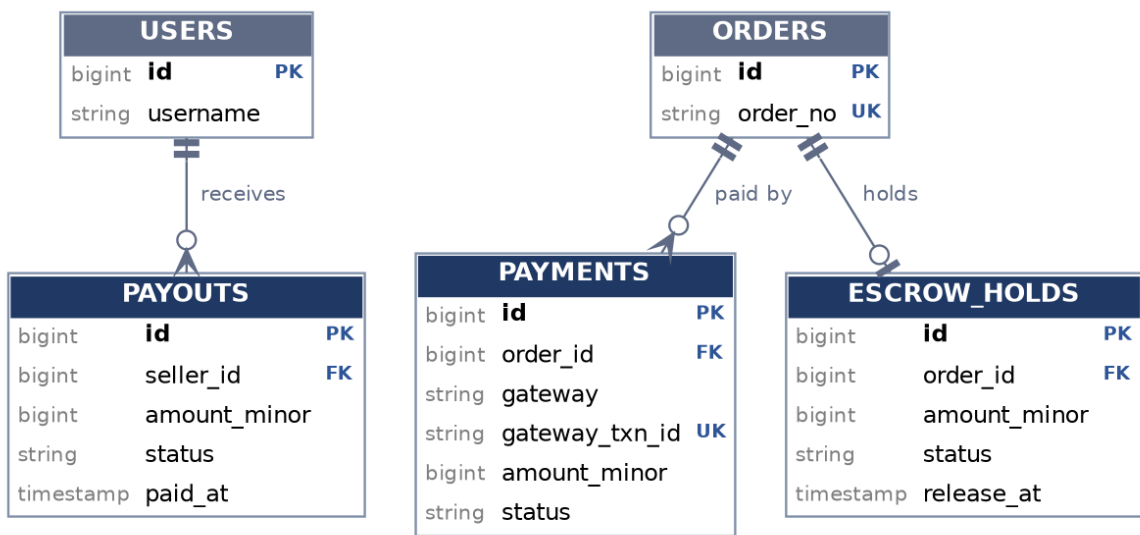


Figure 7: Payments, escrow and seller payouts

Diagram 6: Disputes and ledger

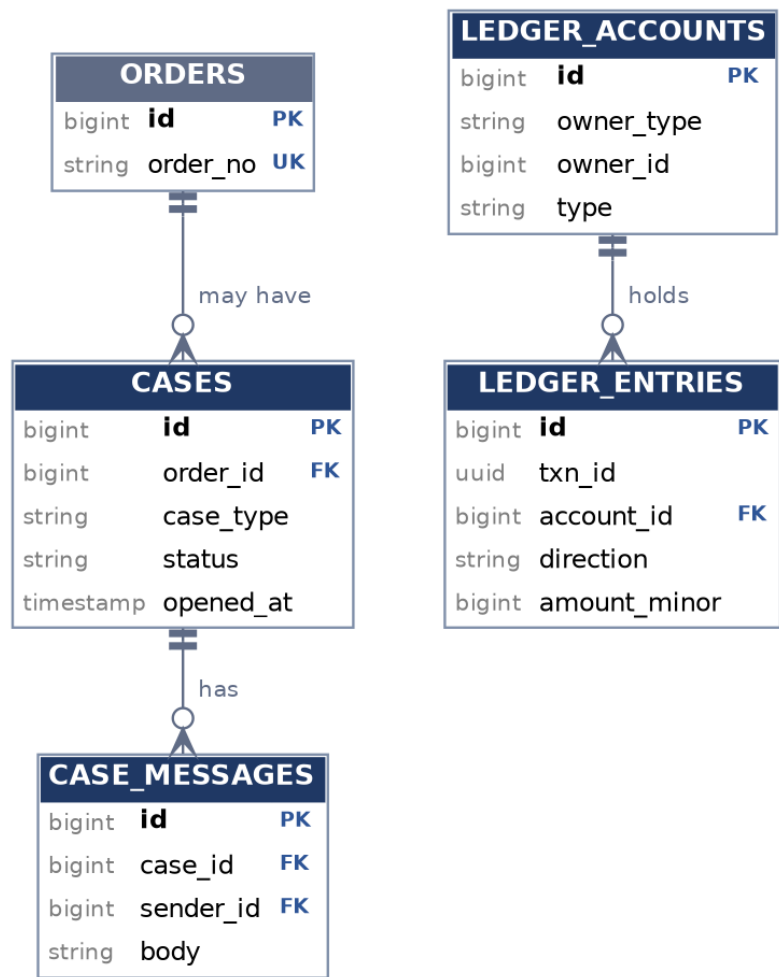


Figure 8: Cases, case messages and the double-entry ledger

Tables not drawn to keep the diagrams clear (carts, coupons, conversations, messages, notifications, fee rules, admin and system tables) connect to users, listings or orders in the same way and are fully listed in section 5.

10. System design and key flows

10.1 System design diagram

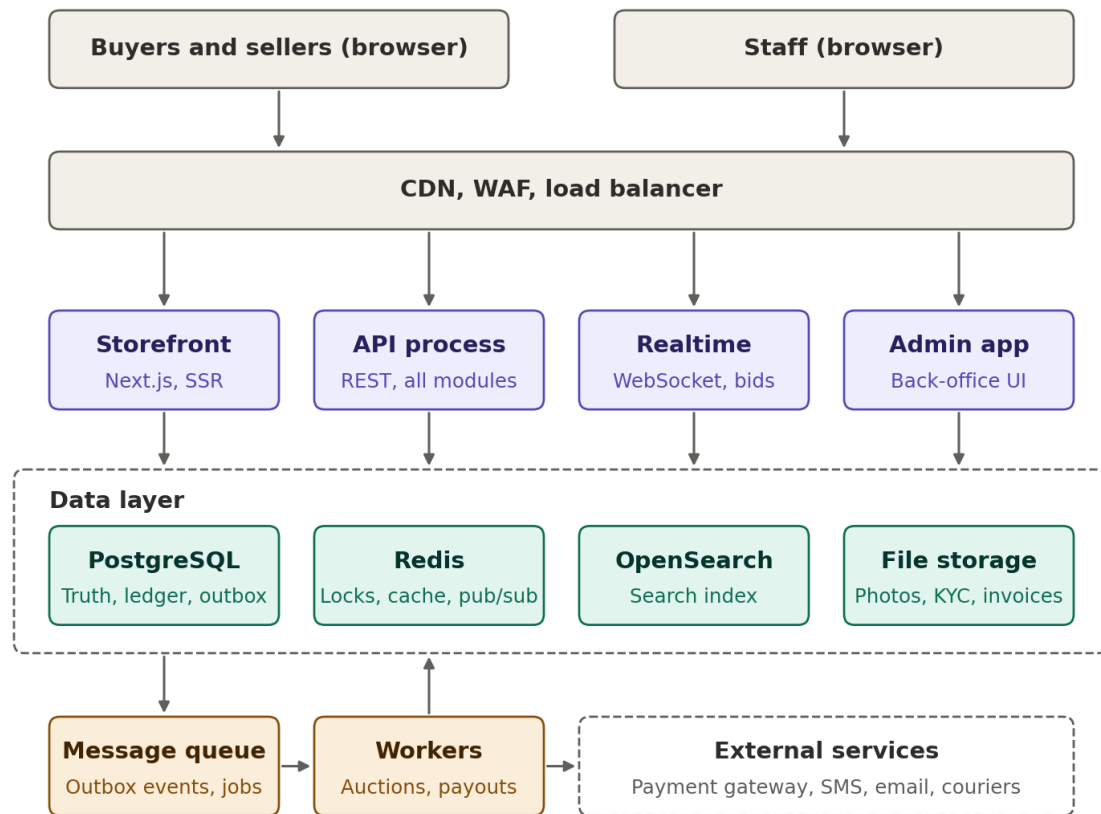


Figure 9: System design of the web-only platform

- **Synchronous path:** browser, then CDN and load balancer, then storefront or API. All writes go through the API process.
- **Live bids:** the API saves the bid, publishes it through Redis, and the real-time process pushes the new price to every watcher within one second.
- **Asynchronous path:** each state change writes an outbox row; a relay publishes it to the queue; workers update search, send notifications, close auctions, release escrow, run payouts and call external services.
- **External services** (payment gateway, SMS, email, couriers) are called only from the API and workers, never from the browser. Gateway webhooks return through the API.

10.2 Key flows

Placing a bid

1. The browser sends the bid with an idempotency key.
2. The API takes a short Redis lock on the auction and opens a database transaction that locks the auction row.
3. It validates: auction running, before end time, bidder is not the seller, bidder not blocked, amount at least current price plus increment.
4. It applies the proxy-bid rule (section 11.1), stores the bid, updates the auction, applies soft close and writes outbox events (new bid, outbid, reserve met).
5. After commit, the new state is published to Redis and pushed to all watchers.

Closing an auction

1. A delayed job fires at the end time; a sweeper runs every 30 seconds as a safety net.
2. If a leader exists and the reserve is met, the auction is marked sold, an order is created as pending payment with a payment deadline (default 48 hours), and both parties are notified.
3. If the buyer does not pay in time, the order is cancelled, a strike is recorded and a second-chance offer can be sent.

Checkout, escrow and payout

1. Stock is reserved atomically for 15 minutes and the order is created as pending payment.
2. A payment is created at the gateway; the buyer pays on the gateway's hosted page.
3. The gateway webhook (signature verified) is the source of truth; on success the ledger moves the amount to escrow and the seller is notified.
4. The seller ships and enters tracking; the buyer confirms delivery, or the protection window passes.
5. If no case is open, escrow is released: seller wallet, platform fee and tax are credited. An open case freezes the release.
6. The payout job pays sellers on their schedule after finance approval.

Listing to search

Publishing a listing saves the row and an outbox event in one transaction; a worker builds the search document and updates OpenSearch. The listing becomes searchable within 2 seconds, and the index can be rebuilt from the database at any time.

10.3 Consistency rules and performance targets

Concern	Mechanism
Two bids at the same moment	Redis lock plus database row lock
Overselling stock	Conditional atomic update of available quantity
Duplicate submits and webhook retries	Idempotency keys; stored gateway event IDs
Lost events	Outbox table with a relay; consumers are idempotent
Concurrent edits	Version column; stale writes rejected
Clock differences	Server time only; client countdown is display-only
Order status	Explicit state machine; illegal transitions rejected and logged

Metric	Target
Availability	99.9% per month
API response (95th percentile)	under 300 ms
Bid placement (95th percentile)	under 200 ms
Search (95th percentile)	under 400 ms
Bid update to watchers	under 1 second
Search index delay	under 2 seconds
Backup recovery point / time	5 minutes / 1 hour

11. Methods and algorithms

11.1 Proxy bidding

Each bidder enters a maximum. The system bids on their behalf, only as high as needed:

- First bid: price is the starting price; the bidder becomes leader.
- Leader raises own maximum: price unchanged, maximum updated.
- New bidder's maximum is higher than the leader's: price becomes the leader's maximum plus one increment (not above the new maximum); the new bidder leads.
- Equal maximums: the earlier bidder keeps the lead at that price.
- New bidder's maximum is lower: the leader stays; price becomes the new maximum plus one increment (not above the leader's maximum).
- Soft close: a bid in the last minutes extends the end time, up to a configured limit.

11.2 Other core methods

Method	How it works
Stock reservation	Atomic conditional decrement of available quantity; reservation released after 15 minutes without payment
Fee calculation	Final value fee = category rate × (price + shipping) + fixed fee, adjusted by seller level and capped; the rule version is stored on each order
Order state machine	Allowed transitions only: pending payment, paid, processing, shipped, delivered, completed; branches for cancel, return, refund and dispute
Escrow release	On delivery confirmation or after the protection window, unless a case is open
Search ranking	Text relevance combined with seller quality, sales velocity, price competitiveness, free shipping, freshness and penalties for policy issues
Fraud checks	Rule engine: account age versus order value, velocity of bids and orders, blacklists, device reuse, unusual bidding on one seller; result is approve, review or block
Idempotency	Payments, bids, orders and offers require an idempotency key; repeated requests return the stored result

11.3 Background jobs

Job	Schedule
Auction sweeper (plus delayed job at end time)	Every 30 seconds
Scheduled listing start; stock reservation release	Every minute
Unpaid order handler; ending-soon and price-drop alerts	Every 5 minutes
Escrow release	Every 15 minutes
Order auto-complete; saved-search alerts; exchange rate refresh (Phase 2)	Hourly
Payout run	Daily
Ledger integrity check; search reindex repair; report generation	Nightly
Seller level evaluation	Monthly

12. Server, storage and payment (client side)

As agreed, the **servers, storage and payment system are provided and paid for by the client**. NepaTronix designs, configures and deploys the application on the infrastructure the client provides. This section defines exactly what is needed so the client can prepare it in time.

12.1 Responsibility split

Item	Provided and paid by	Our part
Application, database and search servers	Client	Specify, configure, deploy, tune
Object storage, CDN, backups	Client	Integrate, set lifecycle rules, test restore
Domain name and SSL certificate	Client	Configure and secure
Payment gateway merchant accounts	Client	Integrate up to two gateways with test and live keys
SMS and email service accounts	Client	Integrate one of each and build templates
Software, code, documentation	NepaTronix	Deliver as described in section 16

12.2 Server requirements (starting point)

Recommended for launch with up to about 50,000 daily visitors and 100,000 active listings. Final sizes are confirmed after load testing in month 4. Cloud or local data-centre hosting is acceptable if it supports Linux virtual machines and managed or self-managed services.

Server	Qty	Size	Purpose
Application (storefront and API)	2	2 vCPU, 4 GB RAM each	Serve web pages and REST API behind a load balancer
Realtime	1	2 vCPU, 4 GB RAM	WebSocket for live bids and chat
Worker	1	2 vCPU, 4 GB RAM	Queue jobs, auction closing, payouts
PostgreSQL primary	1	4 vCPU, 16 GB RAM, SSD	Main database
PostgreSQL replica	1	4 vCPU, 16 GB RAM, SSD	Read replica and failover
Redis	1	2 GB RAM	Cache, locks, pub/sub
OpenSearch	1–2	4 GB RAM each, SSD	Search index
Message queue	1	2 vCPU, 2 GB RAM	RabbitMQ
Staging environment	1	4 vCPU, 8 GB RAM	Testing and client review

Operating system: Ubuntu Server LTS. Servers must keep accurate time (NTP) because auction end times depend on it, and must allow SSH access for the deployment team.

12.3 Storage requirements

Storage	Requirement
Object storage for images and files	S3-compatible bucket, initial 500 GB to 1 TB; about 9 MB per listing with all image sizes; a separate private bucket for KYC documents and invoices
Database disk	SSD, at least 200 GB, expandable
Backups	Daily snapshots and continuous log archiving to a different location; 30 days retention
CDN	Recommended in front of the public image bucket

12.4 Payment system

The platform integrates with payment gateways chosen and contracted by the client (for example wallet or bank gateways used in Nepal). Card and wallet details are entered only on the gateway's hosted page and never reach our servers.

- The client provides merchant accounts, sandbox and live API keys, and webhook access for up to two gateways.
- Payment confirmation relies on verified gateway webhooks, not on the browser redirect.
- The platform records the money flow in its own ledger (payment, escrow, fee, payout, refund) and provides daily reconciliation reports against gateway statements.
- Seller payouts are approved by finance staff in the admin panel and paid out through the client's bank or gateway arrangements.
- Gateway fees and charges are borne by the client.

Regulatory note: holding buyer money in escrow and paying it to sellers may require approvals under Nepali financial regulations. The client is responsible for confirming these requirements (including Nepal Rastra Bank rules) before escrow is used with real customers. The software supports switching escrow off if needed.

13. Security and compliance

- Passwords hashed with argon2id; login rate limiting and lockout; two-factor authentication for sellers and staff.
- TLS on all traffic; encryption at rest; field-level encryption for ID numbers and bank details.
- Card data never stored or processed by the platform (hosted payment pages).
- KYC documents in a private bucket, opened only through short-lived links, every access logged.
- Role-based access for staff; every administrative change is written to the audit log.
- Uploads validated by type and size, location data removed from photos, images moderated.
- Web application firewall and bot protection; dependency and secret scanning in the build; security review before launch.
- Privacy: user consent records, data export and deletion on request, in line with applicable Nepali law.

14. Project plan (4 months)

The project runs for 16 weeks in four monthly milestones. Each milestone ends with a demonstration on the staging site and a written acceptance.

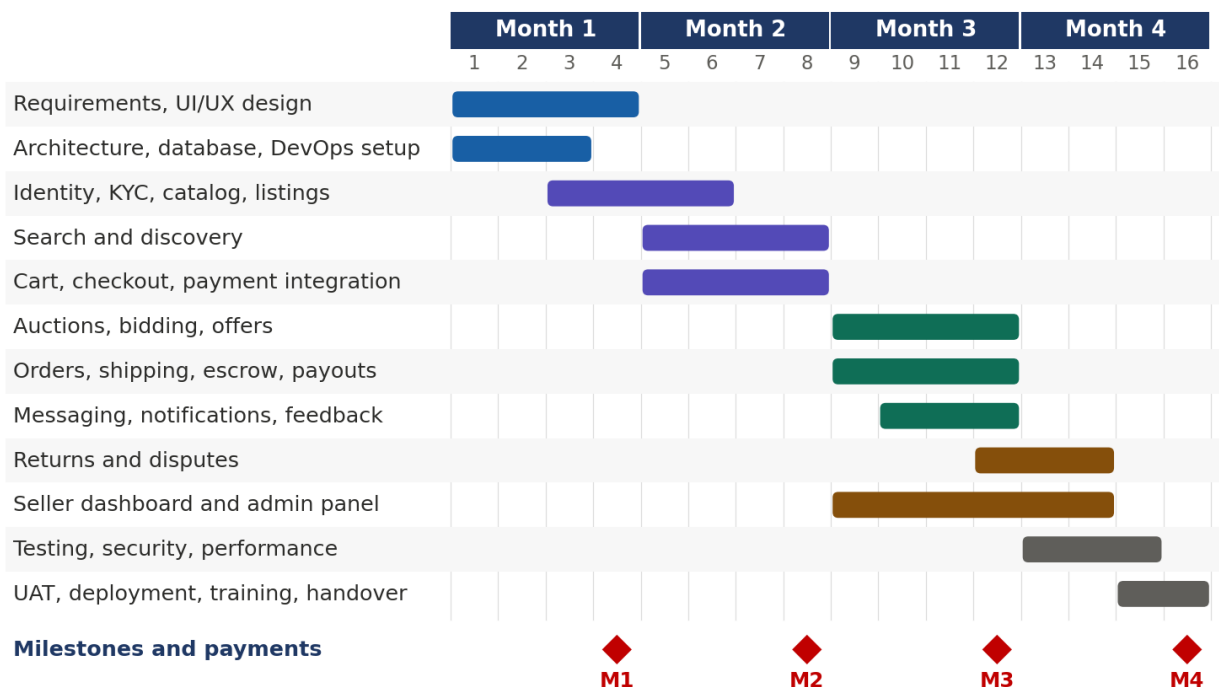


Figure 10: Delivery schedule and milestones (M1 to M4)

Milestone	Deliverables
M1 – End of month 1	Approved requirements and UI/UX designs; architecture and database design; environments and pipeline set up; user accounts, KYC and category catalog working
M2 – End of month 2	Listings (fixed price, photos, drafts, bulk upload); search and filters; cart and checkout; payment gateway integration on staging
M3 – End of month 3	Auctions with proxy bidding and live updates; Best Offer; orders and shipping; ledger, escrow and payouts; messaging, notifications and feedback
M4 – End of month 4	Returns and disputes; seller dashboard and admin panel; testing, security review, performance tuning; user acceptance testing; go-live, training and handover

14.1 Team

Project manager (part-time), technical lead and architect, two backend developers, one frontend developer, one UI/UX designer (part-time), one QA engineer and a DevOps engineer (part-time). A weekly progress call and a demonstration at the end of every two weeks keep the client informed.

15. Cost and payment schedule

Total project fee: NPR 13,00,000 excluding VAT

VAT at the prevailing rate (13%) is added to each invoice. At 13%, VAT is NPR 1,69,000 and the total payable is NPR 14,69,000.

15.1 Cost distribution by work package

Ref	Work package	Amount (NPR)	Share
A	Requirements, UI/UX design, system and database design	1,30,000	10.0%
B	Identity, KYC, catalog and listings	1,50,000	11.5%
C	Search and discovery	80,000	6.2%
D	Cart, checkout and payment gateway integration	1,30,000	10.0%

Ref	Work package	Amount (NPR)	Share
E	Auctions, proxy bidding, offers and real-time updates	1,90,000	14.6%
F	Orders, shipping, returns and disputes	1,40,000	10.8%
G	Ledger, escrow, fees and payouts	1,20,000	9.2%
H	Messaging, notifications and feedback	70,000	5.4%
I	Seller dashboard and admin panel	1,30,000	10.0%
J	Testing, security review and performance tuning	70,000	5.4%
K	Deployment, documentation, training and handover	50,000	3.8%
L	Project management and 30-day post-launch support	40,000	3.1%
	Total (excluding VAT)	13,00,000	100.0%

15.2 Payment schedule

No.	Payment due	Share	Amount (NPR)	VAT 13% (NPR)	Total (NPR)
1	Advance on signing the agreement	25%	3,25,000	42,250	3,67,250
2	Milestone 1 accepted (end of month 1)	20%	2,60,000	33,800	2,93,800
3	Milestone 2 accepted (end of month 2)	20%	2,60,000	33,800	2,93,800
4	Milestone 3 accepted (end of month 3)	20%	2,60,000	33,800	2,93,800
5	Milestone 4: go-live and handover	15%	1,95,000	25,350	2,20,350
	Total	100%	13,00,000	1,69,000	14,69,000

Invoices are payable within 7 days of issue. A milestone is deemed accepted if the client raises no written defect within 5 working days of the demonstration.

15.3 Not included in the fee

- Servers, storage, CDN, backups, domain and SSL (client side, section 12).
- Payment gateway setup fees, transaction charges, SMS and email charges, and any other third-party subscriptions.
- Native mobile apps and all features marked Phase 2.
- Data entry, content writing, photography and legal or regulatory advice.
- Changes requested after approval of the requirements (handled by written change request with a separate estimate).

16. Deliverables and acceptance

16.1 Deliverables

- Working storefront and admin panel deployed on the client's servers.
- Complete source code in the client's repository, with build and deployment scripts.
- Database schema, ER diagrams and migration scripts.
- API documentation (OpenAPI) and architecture document.
- Administrator manual, seller and buyer guides.
- One training session for the client's staff (up to 4 hours) and 30 days of post-launch support for defects.

16.2 Acceptance criteria

- Every Included feature works as described on the staging site during the milestone demonstration.
- No open critical or high-severity defects at go-live.
- Performance targets in section 10.3 are met in a load test on the client's production-sized environment.
- Automated tests for bidding, ledger and checkout pass.

17. Client responsibilities

Item	Needed by
Nominated decision-maker; feedback within 3 working days, milestone acceptance within 5 working days	Start of project
Business rules: categories, fee rates, policies, terms and conditions, logo and brand colours	End of week 4
Staging server and repository access	Week 2
Payment gateway merchant accounts with sandbox keys; SMS and email accounts	Week 4
Production servers, storage, domain and SSL	Week 12
Live payment gateway credentials and regulatory approvals for escrow	Week 13
Test users for user acceptance testing	Week 15

Delays in these items move the delivery date by the same period.

18. Risks and assumptions

Risk	Impact	How we manage it
Expectation of “exactly like eBay”	Scope growth beyond four months	Fixed feature list in section 4; extra features handled as Phase 2 or change requests
Payment gateway approval or API delays	Late checkout testing	Start gateway onboarding in week 1; build against sandbox and a mock gateway
Regulatory approval for escrow	Escrow cannot go live	Escrow can be switched off; client confirms requirements early
Servers not ready on time	Late deployment and load testing	Dates in section 17; staging environment from week 2
Heavy bidding at auction end	Slow or incorrect bids	Locking design, concurrency tests, load test before go-live
Late feedback or changing requirements	Schedule slip	Weekly call, biweekly demo, written change requests

19. Terms and conditions

- **Validity:** this proposal is valid for 30 days from the date shown on the cover.
- **Taxes:** all amounts exclude VAT, which is charged at the prevailing rate.
- **Ownership:** source code and deliverables transfer to the client on receipt of full payment; open-source components keep their own licences.
- **Confidentiality:** both parties keep the other's business information confidential.

- **Change requests:** any work outside the approved scope is estimated and agreed in writing before it starts.
- **Support:** 30 days of defect correction after go-live is included; later support and hosting management can be agreed separately.
- **Third parties:** we are not responsible for outages, fees or policy changes of the client's hosting, payment, SMS or email providers.

20. Approval

By signing below, both parties accept the scope, schedule, fee and terms in this proposal.

For the client

For NepaTronix Engineering Solutions

Name and title

Name and title

Signature

Signature

Date

Date